MILITARY
SPECIFICATION

MIL-M-55029(SigC)
23 July 1958

MONITOR, TELEGRAPH, MULTICHANNEL TH-19( )/FG

1. SCOPE

1.1 This specification covers the requirements for the construction of a telegraph monitor which will monitor data of the 7.42 unit start-stop teletype signal code at speeds of 60, 75 or 100 words perminute and the 7.0 unit code at 65 words per minute, from either a 20 ma or 60 ma neutral loop on up to ten channels simultaneously.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

T-T-271	Twine, Cotton, Wrapping
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-C-843	Cushioning Material, Cellulosic
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant

Military

MIL-P-75	Packaging, Packing and Container-Marking of Electron Tubes
MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-T-152	Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-P-11268	Parts, Materials, and Processes Used in Signal Corps Equipment
MIL-I-11748	Interference Reduction for Electrical and Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-E-15090	Enamel, Equipment, Light-gray (Formula No. 11)

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

DRAWINGS

Signal Corps

SC-C-33073	Form to be Used for Pilot Pack Process Sheet

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification No. A-3 and No. 14

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N.W., Washington 6, D.C.)

UNIFORM FREIGHT CLASSIFICATION

Uniform Freight Classification No. 4

(Applications for copies should be addressed to Uniform Freight Classification, 202 Chicago Union Station, Chicago 6, Illinois.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules No. 21

(Application for copies should be addressed to Consolidated Classification Committee, 1 Park Avenue and 33rd Street, New York 16, New York.)

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3. REQUIREMENTS

3.1 Procurement model.- The procurement model shall be the one which will be available for inspection by prospective bidders and will be lent to the contractor, in accordance with the invitation for bids. Unless otherwise specified herein or in the invitation for bids, the equipment shall conform to the following:

3.1.1 Construction.- Physical construction of the equipment shall conform to the procurement model.

3.1.2 Features.- The equipment shall incorporate the features of the procurement model.

3.1.3 Verification.- As soon as the procurement model is received, with the minimum necessary adjustment and alinement (if any), the contractor shall measure its performance for compliance with specified characteristics of the equipment on the contract. If the model does not meet the specified performance characteristics, the contractor shall adjust or aline it for optimum performance and then repeat the measurements. The contractor shall immediately prepare copies of a test report on the measurements, and forward them as required by the contract. In case of conflict between specified performance characteristics and performance of the model, the former shall govern.

3.2 Parts, materials, and processes, general.- In addition to any requirements of this specification, covering parts, materials, and processes, such items shall conform to Specification MIL-P-11268 including the selection requirements therein.

3.2.1 Marking.-

3.2.1.1 General.- Marking shall conform to Specification MIL-M-13231.

3.2.1.2 Visibility.- Wherever practicable parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum dis-assembly of the equipment.

3.2.1.3 Serial numbers.- The equipment shall be serial numbered.

3.2.2 Tropicalization.-

3.2.2.1 Treatment of assembled equipment.- Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc.) which is applicable on contract.

(2) It does not incorporate any material listed in 3.2.2.2 unless the material has been given the treatment specified in Specification MIL-P-11268 or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer, for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

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3.2.2.2 Materials not resistant to moisture and fungi.- The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.

(b) Cotton.

(c) Cork.

(d) Felt, hair or wool.

(e) Jute.

(f) Leather.

(g) Linen.

(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.

(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.2.2.3 Statement of treatment.- The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. This statement shall include the statements required by Specifications MIL-P-11268 and MIL-T-152, as applicable.

3.3 Finish, protective.- The equipment shall be given protective finish in accordance with Specification MIL-F-14072. The final paint film on type 1 surfaces shall be semigloss light-gray enamel conforming to Specification MIL-E-15090.

3.4 Electrical requirements.-

3.4.1 Frequency coverage.- Each monitor shall be capable of monitoring 7.42 unit code for start-stop teletype speeds of 60, 75 or 100 words per minute and 7.0 unit code at 65 words per minute.

3.4.2 Loop current.-

3.4.2.1 Input.- Each monitor shall be capable of monitoring either 20 ma or 60 ma neutral loops.

3.4.2.2 Current handling capacity.- The input to each monitor shall be capable of withstanding a current overload of 100% on either 20 ma or 60 ma neutral loop without failure of equipment.

3.4.3 Distortion threshold adjustment.- Each of the ten telegraph monitors shall be provided with a detection threshold adjustment permitting the range of peak distortion detection to be varied from 5% of a unit baud to 45% of a unit baud in 5% steps.

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3.4.4 Hit counter threshold adjustment.- Each of the ten telegraph monitors shall be provided with a hit counter threshold adjustment to permit the number of recorded peak distortions which will cause the alarm circuit to be activated to be varied from one to ten.

3.4.5 Automatic reset of hit counter.- Each of the ten telegraph monitors shall have an automatic reset feature associated with the hit counter. The reset circuit will be actuated by reset pulses distributed at 10 minute intervals from a common timing generator. Upon receiving a reset pulse the hit counter shall be automatically reset to the zero condition.

3.4.6. Alarms.-

3.4.6.1 Visual alarm.- Each of the ten telegraph monitors shall be provided with a visual alarm which is actuated when the hit counter reaches the preset number for the individual monitor.

3.4.6.2 Audible alarm.- One audible alarm shall be provided which will be actuated when any of the ten visual alarm circuits are actuated. Means shall be provided to disable this alarm when it is not required.

3.4.6.3 The above alarm circuits shall remain actuated until they are manually reset.

3.4.6.4 Open line or no transition alarm.- A switch shall be provided to select either an open line alarm or a no transition alarm. In the open line alarm position, the visual and audible alarm shall be actuated when there is an absence of input signal (steady-space) for 10 characters or longer. In the no transition position, the visual and audible alarm shall be actuated when there is no transition of the incoming signal for a time equal to 10 characters or longer.

3.4.6.5 Alarm extension circuits.- A terminal strip shall be provided with connections to remote each visual alarm, the common audible alarm and an alarm reset for each monitor.

3.4.7 Calibrate and test features.-

3.4.7.1 Peak distortion meter.- A single distortion calibrate circuit shall be provided, with facilities for connection to each of the 10 telegraph monitors. The peak distortion indicator shall have an accuracy of ±2% at any portion of the scale for the highest signalling speed. Use of this circuit shall not interrupt monitoring of the telegraph channel.

3.4.7.2 Calibrate features for the individual telegraph monitors.-

(a) Each of the ten telegraph monitors shall be provided with means of adjustment of the character timer, the baud times and associated circuits so as to permit attainment of the peak distortion indications with an absolute accuracy of ±2% at each signalling speed, when the adjustments are made in conjunction with the distortion calibrate circuit and a local source of signals, such as a Test Set TS-2( )/TG.

(b) Each telegraph monitor shall also include a manual means of checking the hit counter circuit.

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3.4.7.3 Built-in oscilloscope - The oscilloscope shall be capable of being used as a telegraph distortion analyser as well as providing its ordinary functions.

3.4.7.4 Dot cycle generator - The equipment shall contain a dot cycle generator capable of operation at all the operating speeds of the equipment. It shall be capable of being biased for either space or mark up to 50 percent with an accuracy of 2 percent. The dot cycle generator shall also be capable of keying a telegraph loop.

3.4.8 Power Supply. The equipment shall operate from 115 or 230 volts ± 10%, 50/60 cycle, ± 3 cycle single phase power.

3.4.9 Interference reduction. The equipment shall conform to Class 1 requirements of Specification MIL-I-11748.

3.5 Service conditions. The equipment shall meet the following service conditions:

3.5.1 Equipment nonoperating. The equipment shall comply with the operational requirements of 3.5.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit:

(a) Temperature. Continuous exposure in the range of +160°F to -80°F; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours.

(b) Relative humidity. Relative humidity up to 97 percent, for an indefinite period of time, and exposure at 100 percent relative Humitidy, with condensation, for 4 hours.

(c) Elevation. Elevation up to 25,000 feet above sea level.

(d) Vibration and shock. As described in 3.6.

3.5.2 Equipment operating. The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during world-wide operations.

(a) Temperature. Ambient temperature in the range of +125°F to +25°F. Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours.

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(b) Relative humidity. Relative humidity up to 97 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

(c) Elevation. Elevation up to 10,000 feet above sea leval.

3.6 Vibration and shock.

3.6.1 Vibration. Except for internal resonance of specified parts and sub-assemblies, the equipment shall have no mechanical resonance below 55 cycles per second.

3.6.2 Shock, bench handling. The equipment shall meet full specification performance and show no physical damage when subjected to the test of 4.7.5.

3.7 Preproduction samples. The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract.

3.8 Preconditioning. The equipment shall be capable of meeting the inspection of Section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.9 Interchangeability. Corresponding units and replaceable assemblies, subassemblies, and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used. When dimensions, ratings, characteristics,etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular assembly, subassembly, or part is to be considered replaceable, the contracting officer shall be consulted.

3.10 Technical literature and running spare parts. Technical literature and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

3.11 Workmanship. Workmanship on components covered by this specification shall be such as to meet all the requirements of this specification and any referenced subsidiary specification, drawing, or other document, when inspected in accordance with 4.8, including the applicable portions of the following paragraphs:

(a) In Specification MIL-P-11268:

General requirements for plastic material and plastic parts.	Flux and cleaning agents for soldering.

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Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on Wire and cable.

Process for soldering.
Cleaning of equipment.
Riveting.
General requirements for securing of parts.
Cleaning of welding.
Process for welding.
Gears.

(b) In this specification:

Marking.
Finish.
Tropicalization.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.2)

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3). (See 4.3 and 4.10)

(1) Procurement inspection of items before preparation for delivery. (See 4.3.)

(2) Procurement inspection of preparation for delivery. (See 4.10 and Section 5.)

4.2 Preproduction inspection. This inspection shall consist of the preproduction inspection specified in Table I and group A and group B inspection specified in Tables II and III respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) shock, bench handling; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

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Table I - Preproduction Inspection

Inspection	Rqt. Par.	Insp. Par.
Interference reduction	3.4.9	4.6.12
Temperature	3.5	4.7.1
Moisture resistance	3.5	4.7.2
Altitude	3.5	4.7.3
Vibration	3.6.1	4.7.4
Shock; bench handling	3.6.2	4.7.5

4.3 Procurement inspection of items before preparation for delivery. The minimum inspection performed by the contractor shall be as specified by (a) through (d) below--to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion --will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.4) Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of group A and group B inspection as specified in 4.3.1 through 4.3.2.3.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when it is considered more practical to select separate samples for group B inspection.

(d) Each unit which will be subjected to group A or group B inspection shall be preconditioned after final assembly. (See 3.8).

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4.3.1 Group A inspection. This inspection (including sampling) shall conform to Table II and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group A Inspection

Inspection	Req't Par.	Insp Par.	AQL Major	AQL Minor
Visual and Mechanical	3.11	4.8		
Test Panel Chassis		4.8	1.5 dphu	6.5 dphu
Monitor Chassis (each chassis)		4.8	1.5 dphu	6.5 dphu
(c/o 5 chassis, 2 ea. per chassis)				
Open Line Alarm Chassis		4.8	1.5 dphu	6.5 dphu
Power Supply		4.8	1.5 dphu	6.5 dphu
Relay Rack Cabinet		4.8	1.5 dphu	6.5 dphu
Electrical				
Loop Current	3.4.2.1	4.5		
Input	3.4.2.2	4.6.1		
Current handling capacity	3.4.3	4.6.2		
Distortion threshold adjustment	3.4.4., 3.4.6.1 to 3.4.6.3	4.6.3	1.0 dphu for combined group	*
Hit counter threshold adjustment	3.4.5	4.6.4		
Automatic reset	3.4.6.4	4.6.5		
Open line alarm	3.4.6.4	4.6.6		
No transition alarm	3.4.6.5	4.6.7		
Alarm extension	3.4.7.4	4.6.8		
Dot cycle generator	3.4.7.1	4.6.9		
Peak distortion meter	3.4.7.3	4.6.10		
Oscilloscope				
Operation	3.4.1	4.5	0.65%	
100 WPM input only				

* All electrical defects are considered major

4.3.2 Group B inspection. This inspection (including sampling) shall conform to table III and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of

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nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.4 and the requirements of Standard MIL-STD-105 for disposition of rejected product.

4.3.2.1 Group B sampling plans. The group B sampling plans, as listed in table III, shall be as follows:

Group B plan	AQL	Inspection level for normal inspection	Inspection level for reduced inspection
B-2	4.0%	L-8	L-6

4.3.2.2 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

4.3.2.3 Procedure in case of failure. When an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

Table III - Group B Inspection

Inspection	Req't Par.	Insp Par.	Sampling Plan
Operation			
60,65, and 75 WPM input Signal (operational power)	3.4.1	4.5	B-2 Plan
Over power input range, 50/60 cps ±3 cps; 115/230 vdc ± 10%	3.4.8	4.6.11	B-2 Plan
Interchangeability	3.9.	4.9	B-2 Plan

4.3.3 Reinspection of conforming group B sample units. Unless otherwise specified, sample units which have been subjected to and passed group B inspection may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.4 Disposition of nonconforming product. When defective sample units or failed inspection lots are submitted for acceptance, such product shall

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be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

4.4 Bounce preconditioning. The unit shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skaeneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.5 Operational test. With the loop current set at 20 ma, signals from Test Set TS-2( )/TG shall be fed through the monitor at 60 wpm. With the distortion threshold at 5%, errors should appear on the hit counter. The distortion shall then be removed from the input signal and the hit counter shall not register errors. This procedure shall be repeated at 75 and 100 wpm. To test for 7.0 unit code at 65 wpm, signals from a dot cycle generator shall be used. These tests shall be repeated at 60 ma.

4.6 Electrical tests.

4.6.1 Current handling capacity. Using the set-up of 4.5, the current in the loop shall be doubled (40 ma and 120 ma respectively).

4.6.2 Distortion threshold adjustment. Signals from a Test Set TS-2/TG shall be fed through the monitor. The distortion shall be set 2% above the distortion threshold. Hits shall be recorded. The distortion shall be reduced to 3% below the distortion threshold. Hits shall not be recorded. Testing shall be made at all settings of the distortion threshold.

4.6.3 Hit counter threshold. Distorted signals shall be fed through the monitor to actuate the hit counter. Both the visual and audible alarms shall be actuated when the hit counter reaches the preset number. This shall be done for all settings from 1 to 10.

4.6.4 Automatic reset. Distorted signals shall be fed through the monitor to actuate the counter to a number below the alarm threshold. Both the visual and audible alarms shall remain inactive. The hit counter shall reset to zero ten minutes after the previous reset.

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4.6.5 Open line alarms. The monitor shall be set for "open line" alarm. Signals shall be fed through the monitor and then removed. The open line alarm shall be actuated after a time interval equal to 10 characters.

4.6.6. No transition alarm. The monitor shall be set to the "no transition" alarm. A mixed code shall be fed through the monitor and then an all "space" signal shall be sent. The no transition alarm shall be actuated after a time interval equal to 10 characters. This shall be repeated for an all "mark" signal.

4.6.7 Alarm extension. All alarms and resets shall be extended to a remote location from the provided terminal strip and the alarm tests of 4.6.3, 4.6.4, 4.6.5, and 4.6.6 shall be repeated. The correct alarms shall be seen and heard at the remote point.

4.6.8 Dot cycle generator.

4.6.8.1 Signals from the dot cycle generator shall be fed into a Hewlitt-Packard Counter 524B or equal, and the frequency checked at 60, 65, 75, and 100 wpm, which are frequencies of 22.77, 22.77, 28.4 and 37.1 respectively.

4.6.8.2 Signals from the dot generator shall be fed into Test Set Teletypewriter TS-917/GG and the distortion of the dot cycle generator shall be changed to several points on the meter. The reading on the test set shall be within ±2%.

4.6.8.3 An external telegraph loop shall be set up and signals from the dot cycle generator fed into it. An oscilloscope shall be placed across the loop input and checked for correct signals at 60,65,75 and 100 wpm at both 20 ma and 60 ma.

4.6.9 Peak distortion meter. Signals from the dot cycle generator shall be fed into the peak distortion meter. Various amounts of distortion shall be introduced. The peak distortion meter shall read within ±2% of the input distortion.

4.6.10 Oscilloscope. Signals from the dot cycle generator shall be fed into the oscilloscope and different amounts of distortion introduced. The oscilloscope shall indicate this distortion within ±2% of the input distortion. All controls of the oscilloscope shall be turned to see that they are working properly.

4.6.11 Power regulation. The input power source to the monitor shall have the voltage varied plus or minus 10 percent and the frequency plus or minus 3 cycles. The equipment shall pass the operational test of 4.5 for 115VAC and then at 230VAC.

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(d) Unsatisfactory workmanship evidenced by poor quality of manufacture, improper fitting, finishing inadequate for the physical and

4.6.12 Interference reduction. The equipment shall be tested to determine compliance with 3.4.9.

4.7 Service conditions test.

4.7.1 Temperature. The equipment shall be subjected to the temperature cycle shown on MIL-STD-169. Tests shall be made at step 4 and at step 9 as well as steps required by MIL-STD-169.

4.7.2 Moisture-resistance test.

4.7.2.1 Test conditions.

(a) Do not remove equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.7.2.2 Test procedure. The sample equipments will be tested as follows:

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Measure the performance and readjust if necessary to meet specification requirements.

(d) Subject to continuous cycling for a total of five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. Measure performance in accordance with the periods indicated on drawing for the first four cycles.

(e) After cycling has been completed, condition the equipments for 24 hours at 77° ±5°F at 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. After adjustment, the equipments shall meet the performance specified in 4.7.2.3

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4.7.2.3 Performance. Repeat the tests of 4.7.2.2. There shall be no mechanical deterioration, as evidenced by corrosion of metal parts or binding of rotating parts, and there shall be no degradation in performance from that obtained prior to the five humidity cycles.

4.7.2.4 Failure. If any component fails to meet the performance specified in 4.7.2.3 during cycling, it shall be realined or readjusted once. If the component then fails to meet 4.7.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the component fails to meet full specification requirements after conditioning and adjustment, as specified in 4.7.2.2(e), it does not pass the test.

4.7.3 Altitude test. The equipment shall be placedin an altitude chamber at normal conditions of temperature, pressure, and humidity, and shall be tested to determine compliance with electrical requirements. The pressure shall then be reduced to 20 inches of mercury and stabilized for 2 hours. The equipment shall again be tested. The pressure shall then be lowered to 11 inches of mercury and the chamber maintained at this pressure for another 2-hour period. The pressure shall then be increased to 29.9 inches of mercury and tested again. There shall be no degradation of performance in the second and third measurements from those established in the first test.

4.7.4 Vibration test. The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.6.1 as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. [illegible] The vibration table shall provide approximately sinusoidal vibration.

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

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(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.7.5 Shock test; bench-handling. The chassis-and-front-panel assembly shall be placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit assembly to drop back to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly could be placed practicably during servicing.

4.8 Visual and mechanical inspection. The equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.9 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major components.

(d) Location of connectors, locking pins, fasteners, slides, and mountings on plug-in assemblies.

(e) Size and form of special threads.

4.9.1 Electrical interchangeability. All subassemblies being tested for interchangeability shall be substituted for a like subassembly in an equipment which has been subjected to and met the operational test. The operational test shall then be repeated, without any degradation in performance.

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4.10 Procurement inspection of preparation for delivery. The minimum inspection performed by the contractor shall be as specified in 4.10.1 and 4.10.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government - at its discretion - will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.4.) Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.10.1 Preservation and packaging. Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table IV.

TABLE IV

PRESERVATION, AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Incorrect packaging method applied.	2. Packaging material damaged.
3. Cushioning or padding omitted.	3. Conforming wraps are not snugfitting and contain voids.
4. Cushioning inadequate for the physical and mechanical protection of the item.	4. Any item of marking information other than 6 thru 8 omitted, incorrect, or illegible.
5. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.	
6. Stock number omitted, incorrect, or illegible.	
7. Nomenclature omitted, incorrect, or illegible.	
8. Marking of quantity of items in package omitted, incorrect or illegible.	

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4.10.2 Packing and marking of exterior containers. Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table V.

TABLE V

PACKING, AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Unsealed carton.
2. Quantity in pack not as specified.	2. Defective taping or sealing of carton.
3. Box closure not as specified.	3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Type, grade, class, and style of the shipping container not as specified.	4. Any item of required marking information other than 9 thru 15 listed under major defects omitted, incorrect, or illegible.
5. Strapping omitted.	
6. Strapping inadequate or incorrectly applied.	
7. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.	
8. Shipping documents or packing list omitted.	
9. Stock number omitted, incorrect, or illegible.	
10. Nomenclature omitted, incorrect, or illegible.	
11. Marking of quantity of items in pack omitted, incorrect, or illegible.	
12. Destination marking omitted, incorrect, or illegible.	
13. Service designation (color marking) omitted.	
14. Specified special marking and labeling such as NAP labels shipment digit markings, etc. omitted, incorrect, or illegible.	
15. Overseas code markings omitted, incorrect, or illegible.	

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4.10.2.1 Inspection lot. A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.10.2.2 Procedure in case of failure. If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.10.2.3 Disposition of nonconforming product. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.11 Rough handling test (preparation for delivery). When rough handling test is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:4.5 Operational Test

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Telegraph Monitor Multichannel TH-19/FG shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Technical manual. Each technical manual shall be packaged Method 1C-3 as follows: Inclose each technical manual within a type 1 class b bag conforming to Specification MIL-B-117. Heat seal the bag to effect closure.

5.1.1.2 Tubes. All tubes over four inches in length not securely clamped in place shall be removed and together with spare tubes shall be packaged in accordance with the applicable provisions of Specification MIL-P-75.

5.1.1.3 Accessories and electrical spare parts. Accessories and electrical spare parts, determined in quantity and number from the bid request or contract shall be packaged in accordance with the applicable provisions of Specification MIL-P-116.

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5.1.1.4 Consolidated package. Consolidate the items packaged as specified in 5.1.1.2 and 5.1.1.3 together within a close-fitting regular slotted fiber box, class 1 conforming to Specification PPP-B-636. Box closure shall be in accordance with the appendix of the referenced specification.

5.1.1.5 Telegraph Monitor, Multichannel, TH-19/FG.

5.1.1.5.1 Prepackaging. Cushion the entire front area of the telegraph monitor with a blanket fabricated of cellulosic cushioning material type III, class A conforming to Specification PPP-C-843. Secure the cushioning agent with water resistant pressure sensitive tape conforming to Specification PPP-T-76. The rack mounted components in the rear of the monitor cabinet shall be blocked and braced by means of cells or pads or both, fabricated of double faced corrugated fiberboard having a minimum dry bursting strength of 200 pounds, and designed to prevent free movement of the components and shearing stress on the mountings. Fiberboard shall meet the applicable provisions of Specification PPP-B-636. Place the consolidated package (see 5.1.1.4) within the cushioning cells. The rear door of the cabinet shall be closed and secured.

5.1.1.5.2 Packaging. The prepackaged Telegraph Monitor, Multichannel, TH-19/FG shall be packaged Method 1C-5 as follows: Wind the power cord into a coil of proportionate dimensions and tie at three places with type II cotton twine conforming to Specification T-T-871, securing the free ends of the cord to the coil with the ties. Place the coiled cord within the metal channel on top of the unit and secure. Continue packaging following the techniques illustrated in Figure 1. Before placement of the upper end cap a cell of double faced corrugated fiberboard shall be fabricated of such dimensions as to square up the upper end of the monitor and permit a close-fitting end cap.

5.1.2 Level C. Telegraph Monitor, Multichannel TH-19/FG shall be packaged in accordance with commercial practice, and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 Level A. Telegraph Monitor, Multichannel TH-19/FG shall be packed for shipment within a close-fitting style 2 nailed wood box, conforming to Specification PPP-B-621. The box shall be modified to include a skid type base. The inside length of the box, shall be comparable to the long dimension of the packaged item. Place the technical manual (see 5.1.1.1) between contents and lid of the box. Box closure shall be in accordance with the box specification.

5.2.1.1 Metal strapping. Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

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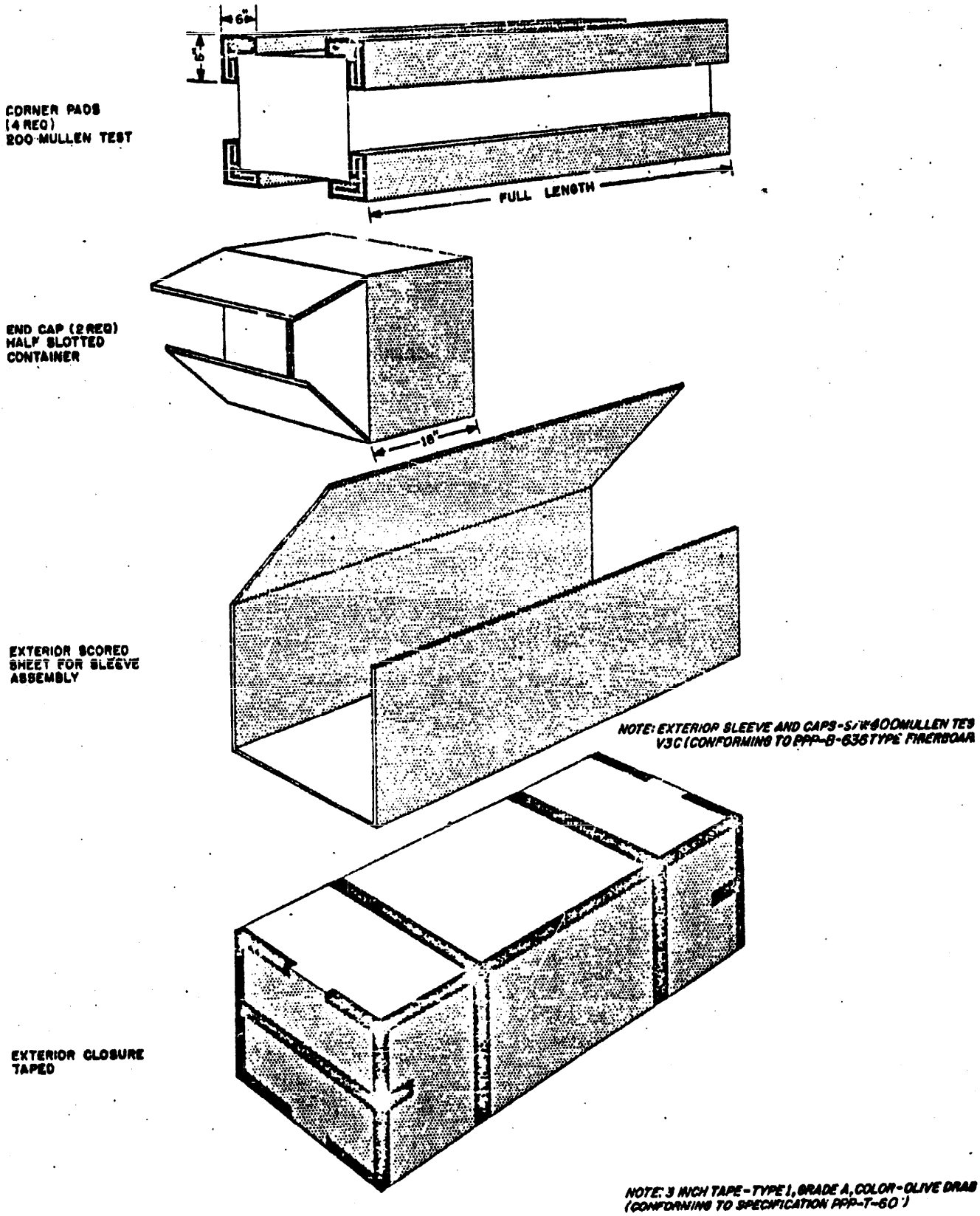


FIGURE 1

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5.2.2 Level B. Telegraph Monitor, Multichannel TH-19/FG shall be packed for shipment as specified in 5.2.1.

5.2.3 Level C. Telegraph Monitor, Multichannel TH-19/FG shall be packed for shipment in a manner conforming to Uniform Freight Classification No.4, and Consolidated Freight Classification Rules 21 for rail shipment, National Motor Freight Classification No.A-3 and No.14 for truck shipment, and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Monitor, Telegraph, Multichannel TH-19/FG is used for detecting errors caused by distortion and circuit fading on from one to ten teletypewriter loops.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level of packaging and level of packing required for shipment. (Level A or level B)

c. The specific paragraphs of section 5 which are applicable to the particular procurement.

d. Preproduction inspection:

(1.) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample.

(2.) Preproduction pack(s) as follows:

(a) Makeup of pack(s).
(b) Number of each kind of pack to be submitted.
(c) Inspection to be performed thereon—including rough handling test, which will not be performed as procurement inspection.
(d) Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

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(e.) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of tools and running spare parts required.

i. Submission of the statement of treatment referenced in 3.2.2.3, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TM-19W/FG. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Verification inspection. The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.5 Definition of level B. Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

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